What a Campus Network Actually Looks Like: Traffic Composition, Concentration and Short-Window Stability

ST5290 Data Science Industry Project,
Recorded presentation (about 9 minutes)

Ruan Yanxin (A0333643H), MSc in Statistics, NUS
Company: Zhejiang SuanWangYun Intelligent Technology Co., Ltd., Data & Artificial Intelligence Center
Company supervisor: Fu Zhiyuan
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The question, and the short answer

- **What we asked:** how is traffic composed, how concentrated is it, and is the platform stable in time?
- **Why it matters:** capacity and monitoring decisions depend on which of those three is true.
- **Short answer:** the bytes come in from outside and sit on very few hosts, and inside the window the platform looks steady.

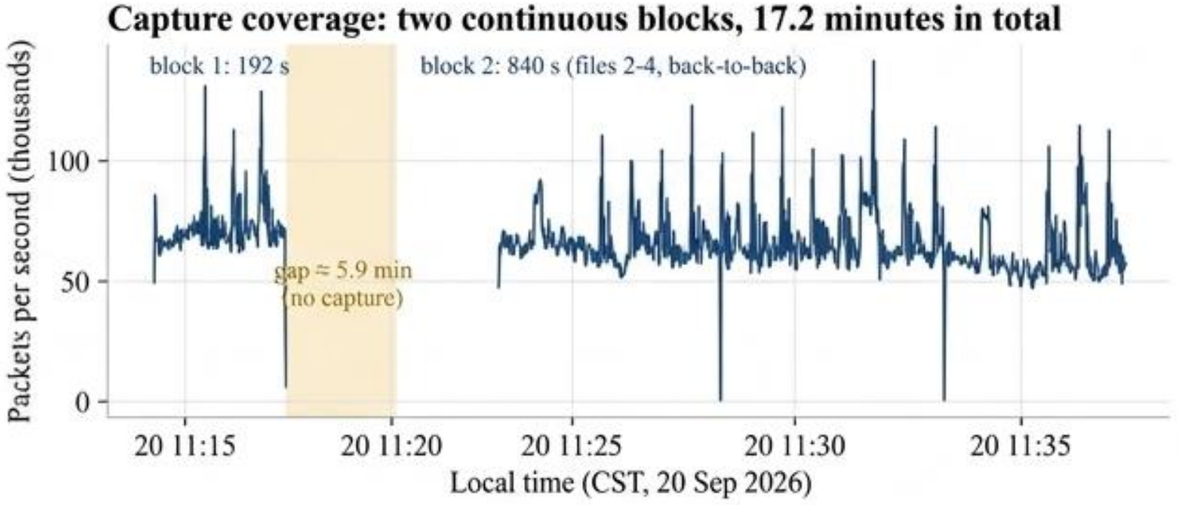
My Task:

- analyse the traffic of the platform's campus network
- turn the result into something the network team can use

Three questions, three kinds of evidence.

What we captured, and what we cannot claim

- **The data:** four capture files from the core forming two continuous blocks, 17.2 minutes in total, 68.7 million packets, 36.2 GB on disk, and 34.85 GB of one-way traffic.
- **The scale:** 49,560 hosts, of which 5,223 are inside and 44,335 are external peers.
- **The honest limit:** one morning only, so no daily or weekly pattern can be claimed.



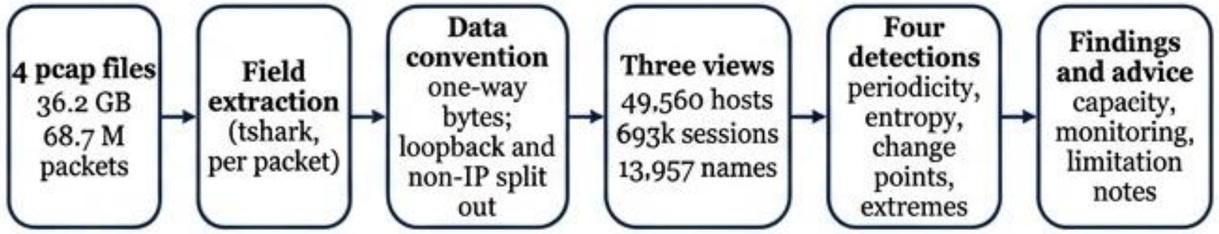
Each point = one second of capture; gaps are seconds with no data. Files 2/3/4 meet end-to-start, so they form one 840 s cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Source: core captures, 17.2 min, two continuous blocks.

A careful snapshot, not a baseline.

How we measured

- One-way byte convention: every packet is counted once, and the classes sum to the total.
- Three analysis units: hosts, sessions which are a bidirectional five-tuple with at least two packets, and segments.
- Four detectors: concentration, port entropy, periodicity, and change points with extreme values, each with a false-discovery control.



Source: core captures, 17.2 min, two continuous blocks.

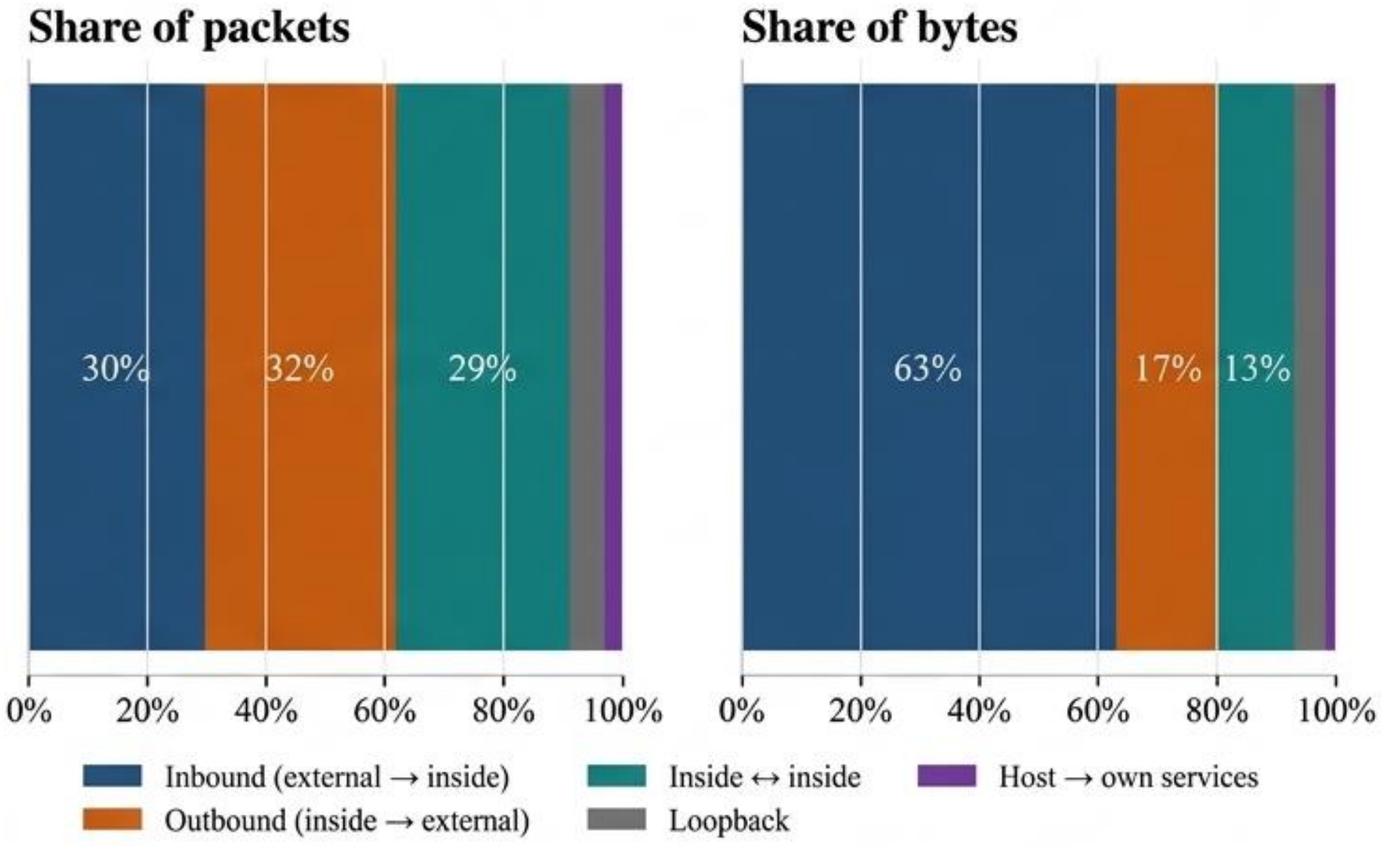
Tools

tshark: field extraction from the raw captures
Python: pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib
Quarto + LaTeX: report and figures

Every number later in this talk comes from one of these three units.

The bytes are inbound, the packets are not

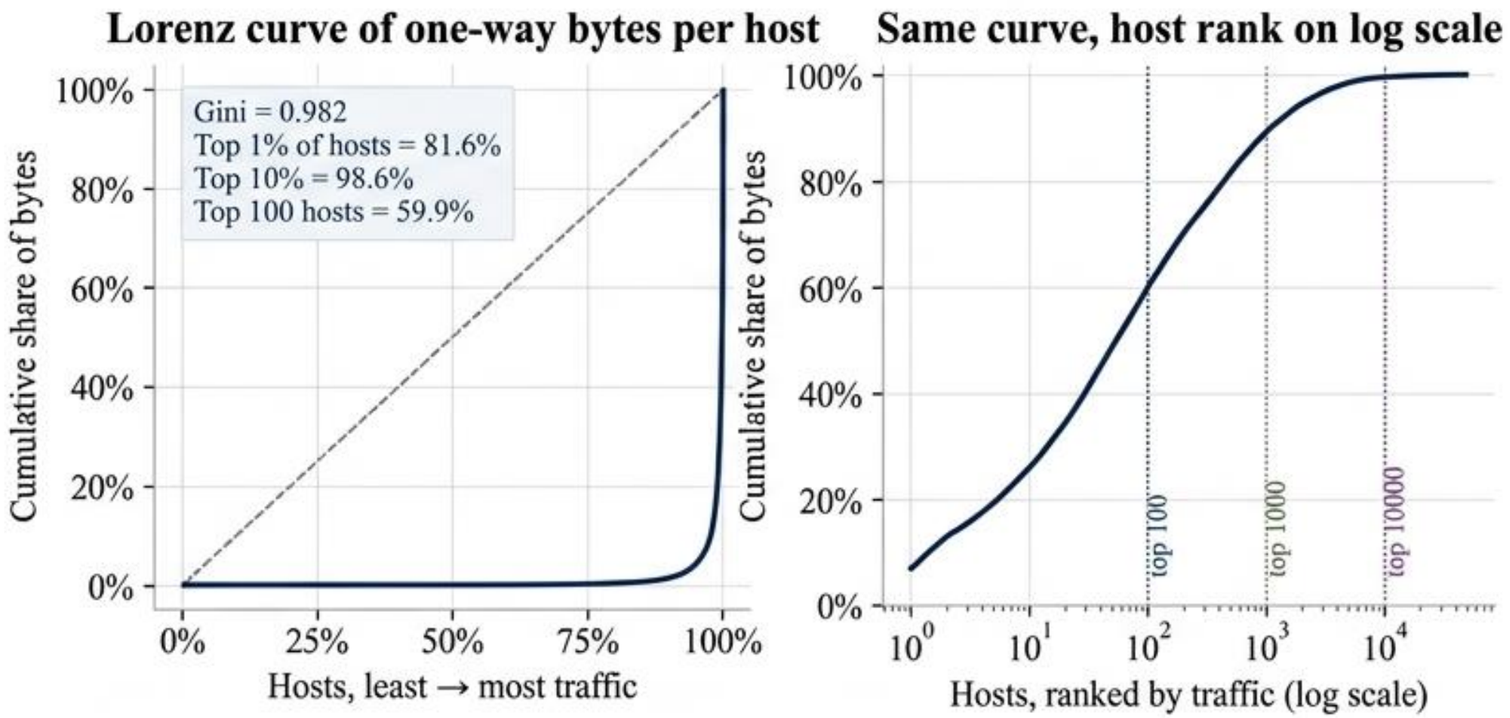
- **Bytes:** inbound traffic is 21.6 GB, or 63.1% of the classified total, while outbound traffic is 5.9 GB, or 17.4%.
- **Packets:** the volume is almost symmetric: the in/out packet ratio is 0.93 while the byte ratio is 3.64.
- **So what:** planning capacity by connection count will underestimate inbound bandwidth pressure.



Source: core captures, 17.2 min, two continuous blocks.

There is no average host

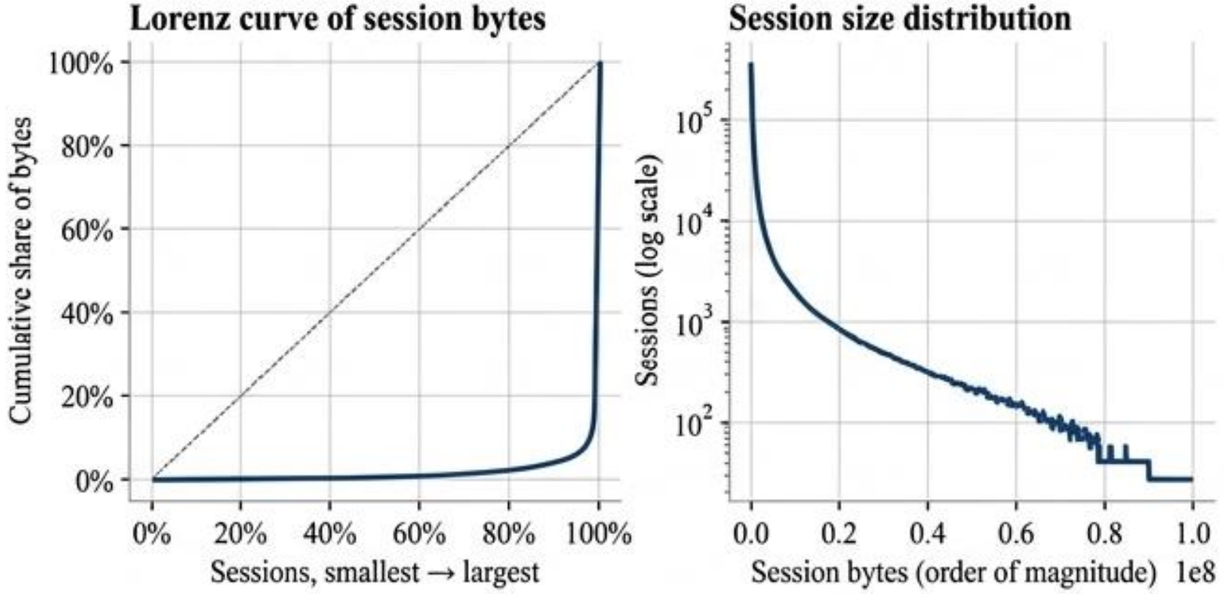
- **Concentration**: the top 1% of hosts carry 81.6% of the bytes, and the top 10% carry 98.6%.
- **A hundred machines**: the top 100 hosts alone account for 59.9% of the total volume.
- **So what**: one network-wide alert threshold cannot fit both the heavy few and the silent majority, so thresholds have to be tiered by group.



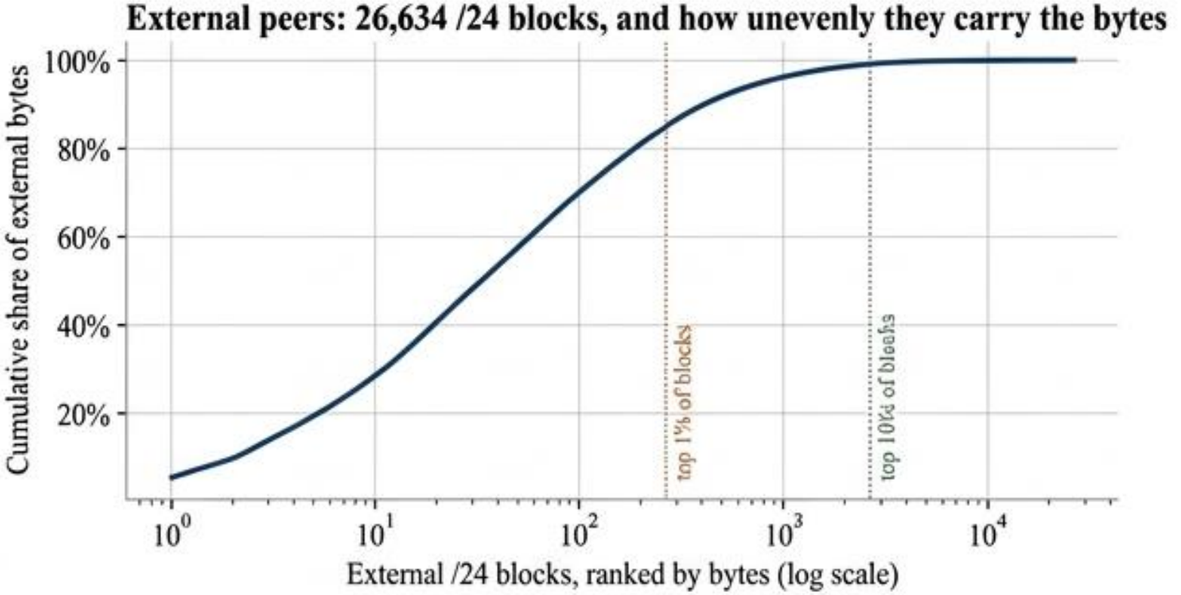
Source: core captures, 17.2 min, two continuous blocks.

The same concentration, twice more

- **Sessions:** across 693,484 sessions, the largest ten carry 10.5% of session bytes and the top thousand carry 66.1%.
- **External peers:** among 26,634 candidate /24 blocks, the top 1% carry 85.0% of external bytes.
- **So what:** concentration is a property of the traffic, not an artefact of one way of slicing it.



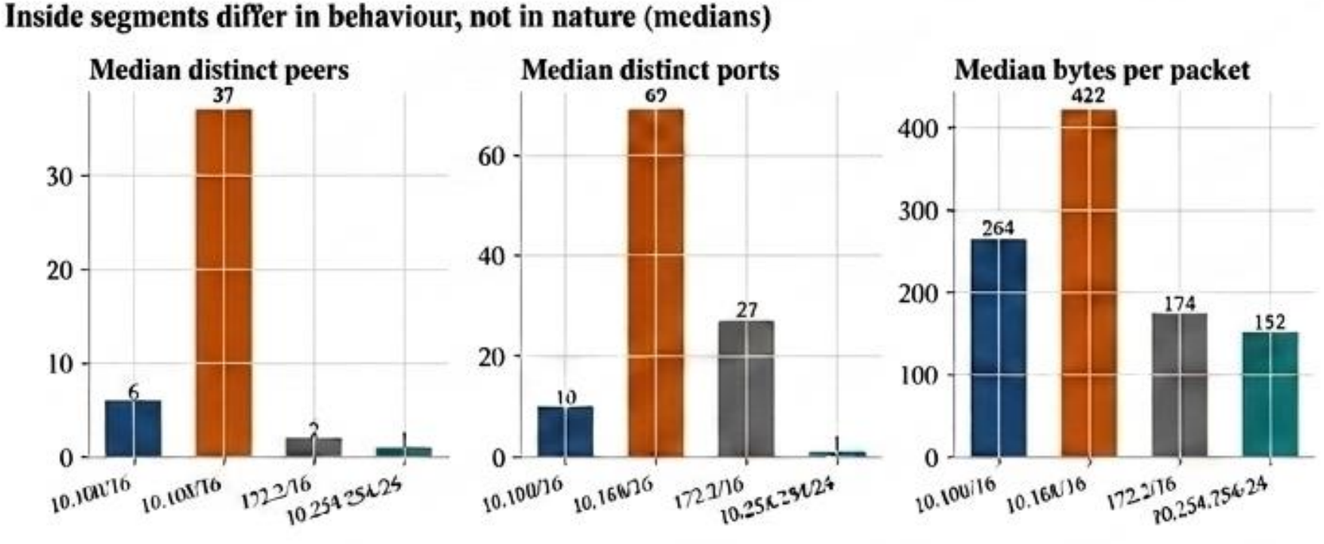
Source: core captures, 17.2 min, two continuous blocks.



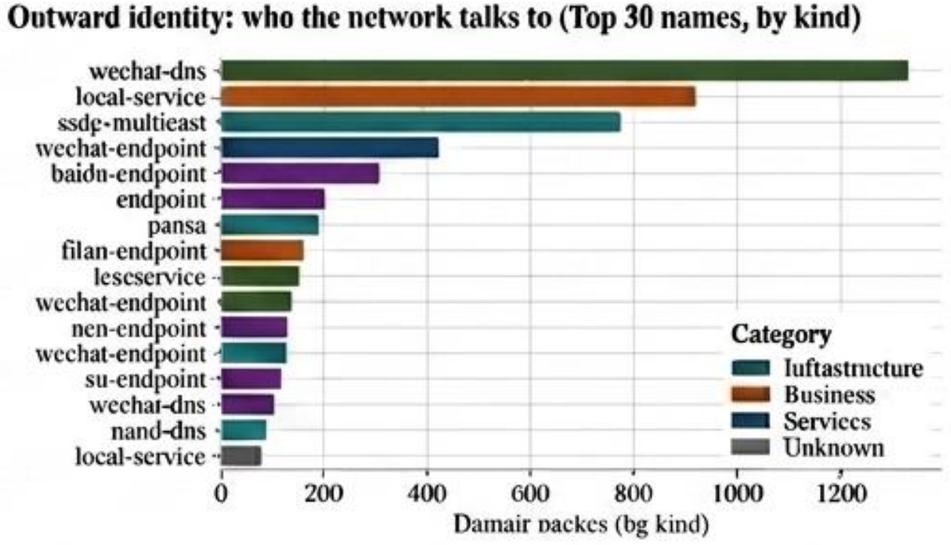
Source: core captures, 17.2 min, two continuous blocks.

Four layers, and who they talk to

- **Four groups**: a transparent rule cuts hosts into groups of 478, 772, 45, and 3,928, while a data-driven clustering on the same features agrees only moderely, with an adjusted Rand index of 0.41.
- **Inside segments**: the 10.100/16 segment behaves like services, 10.168/16 acts like end-user devices, and 172.2/16 looks like infrastructure.
- **Outward identity**: the busiest destinations are the SSDP multicast address, a WeChat endpoint, and local service discovery, rather than obvious workload names.



Source: core captures, 17.2 min, two continuous blocks.

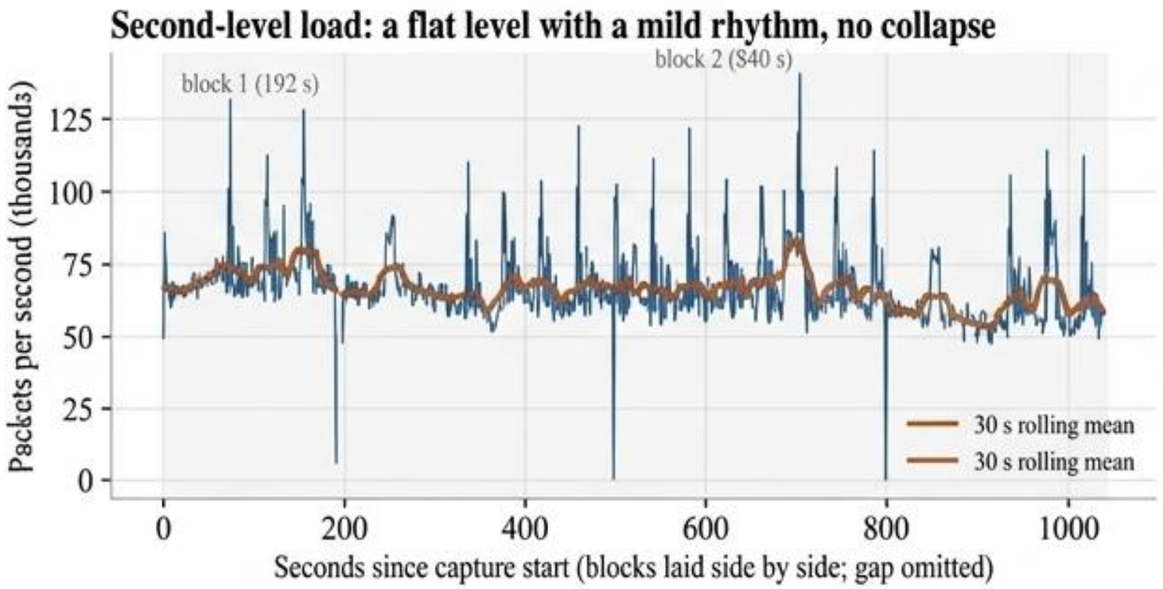


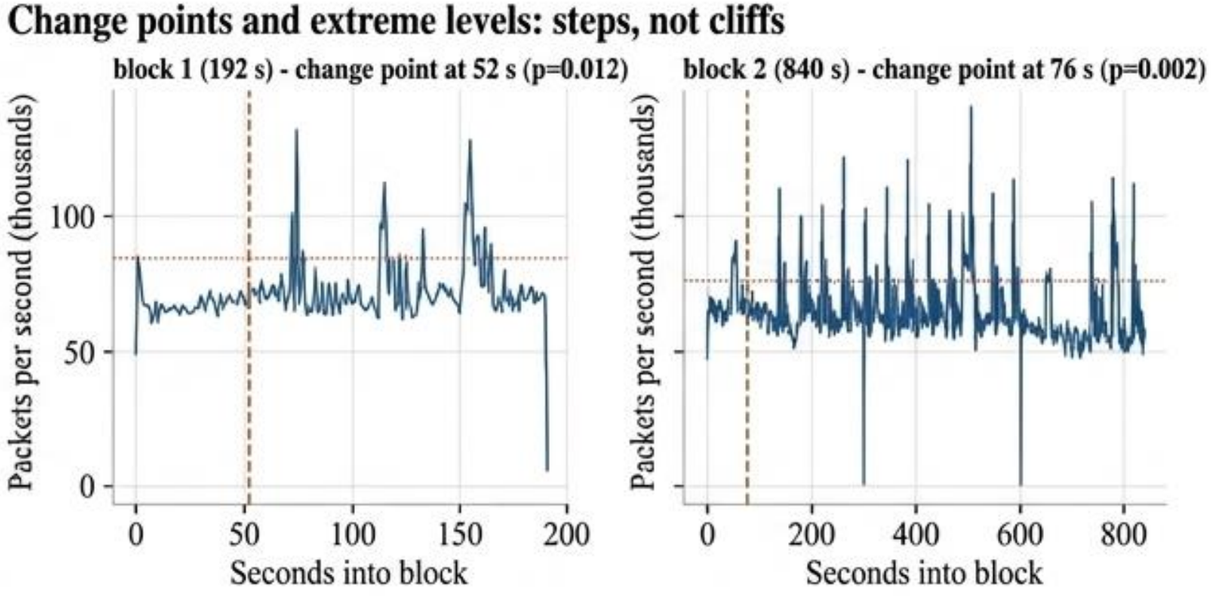
Source: core captures, 17.2 min, two continuous blocks.

The groups are a statistical cut, not yet a business fact.

Stable inside the window, and only inside it

- **Level**: the rate is 71,735 packets per second in the first block and 65,230 in the second, with a coefficient of variation below 0.19.
- **Changes**: there is one step in each block, of 9.5% and 10.2%, with a bounded upper tail.
- **So what**: steady inside these 17.2 minutes, which is precisely why it cannot serve as a baseline.

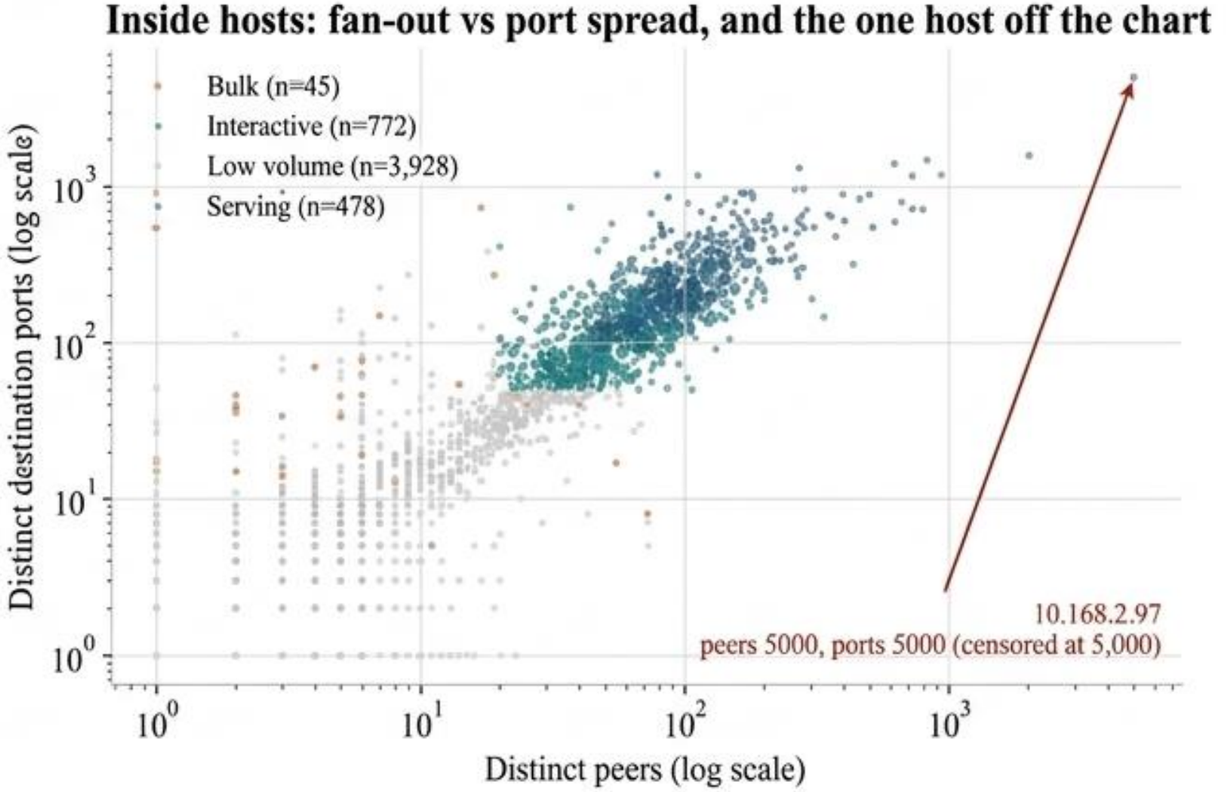




Source: core captures, 17.2 min, two continuous blocks.

Leads worth chasing, not conclusions

- **One host off the chart**: the host 10.168.2.97 reaches the 5,000 measurement ceiling on both peer count and port count.
- **Two more patterns**: there are 159 hosts spread across an unusually wide port range, and two source ports carrying almost identical volume from a single host.
- **So what**: each lead is written as a candidate, with the reason it cannot be confirmed from this data alone.



Source: core captures, 17.2 min, two continuous blocks.

What we recommend, and what we would improve

Recommendations first, then what we would improve next time.

Recommendation	What it unlocks	Cost
Tier monitoring thresholds by host group	One threshold that fits both heavy and quiet hosts	More configuration, periodic reviews
Weight capacity to inbound and elephant flows	Capacity that matches where the bytes actually are	A change of convention, so old numbers are not comparable
Plan connections and bandwidth separately	Two capacity models that stop mixing causes	Two models to maintain
Fix six metrics and report them monthly	Comparable numbers over time	Scheduling and retention only
Capture a full business cycle	Statements about peak hours, not just this window	Storage and a retention policy
Cross-check with device-side counters	Removes the one-way-versus-bidirectional doubt	Device access and clock alignment

Implementing these recommendations shifts the platform from reactive guessing to baseline-driven insights.

What this does and does not say

- **What it does say**: a quantified picture of composition and concentration that survived every robustness check we ran.
- **What it does not say**: who the users are, which applications this is, or how the network behaves day to day.
- **Takeaway**: the network is inbound-dominated and extremely concentrated, so measure again before changing anything.

Thank you. Questions can come to me by email.

e1597848@u.nus.edu